SDG 14 ER Diagram (Rough Draft, Text Version)

Hudson River Water Quality Database, Newburgh NY · MySQL 8
8 entities: 4 core tables (Drone, Run, Reading, WaterSample), 1 reference table (SamplingPoint), 2 lookup tables (OperationalStatus, PolymerType), and 1 junction table (SampleDetection). Key: PK = primary key, FK = foreign key.

1. Entities and attributes

Drone

Field	Data type	Key	Notes
drone_id	INT	PK	Auto-increment
drone_name	VARCHAR(50)		e.g. HudsonScout-01
model	VARCHAR(50)		Optional
commissioned_date	DATE		

Run

Field	Data type	Key	Notes
run_id	INT	PK	One row per trip
drone_id	INT	FK → Drone	
water_body_name	VARCHAR(60)		"Hudson River", stored once per run
start_time	DATETIME		
end_time	DATETIME		NULL while the run is in progress

SamplingPoint

Field	Data type	Key	Notes
point_id	INT	PK	
point_label	VARCHAR(20)	UNIQUE	NB-01 to NB-08
river_mile	DECIMAL(5,2)		Orders points along the river
latitude	DECIMAL(9,6)		
longitude	DECIMAL(9,6)		
nearest_runoff_source	VARCHAR(80)		NULL if none
runoff_distance_m	INT		Metres

OperationalStatus (lookup)

Field	Data type	Key	Notes
status_code	VARCHAR(30)	PK	ACTIVE_SAMPLING, SENSOR_FAULT, …
description	VARCHAR(150)		

Reading

Field	Data type	Key	Notes
reading_id	INT	PK	One row per stop
run_id	INT	FK → Run	UNIQUE together with point_id
point_id	INT	FK → SamplingPoint	UNIQUE together with run_id
status_code	VARCHAR(30)	FK → OperationalStatus	
recorded_at	DATETIME		
latitude	DECIMAL(9,6)		Actual GPS position
longitude	DECIMAL(9,6)		
water_temp_f	DECIMAL(5,2)		CHECK 32–100; NULL if the probe fails
ph	DECIMAL(4,2)		CHECK 0–14
dissolved_oxygen_mgl	DECIMAL(5,2)		CHECK ≥ 0
turbidity_ntu	DECIMAL(6,2)		CHECK ≥ 0
salinity_ppt	DECIMAL(5,2)		CHECK ≥ 0
flow_rate_ms	DECIMAL(5,2)		Can be negative (tidal river)
hazard_flag	BOOLEAN		Default FALSE

WaterSample

Field	Data type	Key	Notes
sample_id	INT	PK	One row per bottle
reading_id	INT	FK → Reading, UNIQUE	UNIQUE makes it one-to-one
collected_at	DATETIME		
volume_l	DECIMAL(4,2)		Default 1.00
analyzed_at	DATETIME		NULL until the lab analyzes it

PolymerType (lookup)

Field	Data type	Key	Notes
polymer_id	INT	PK	
polymer_name	VARCHAR(60)	UNIQUE	e.g. Polyethylene
abbreviation	VARCHAR(10)	UNIQUE	PE, PP, PET, PS, PVC

Field	Data type	Key	Notes
common_sources	VARCHAR(120)		

SampleDetection (junction)

Field	Data type	Key	Notes
sample_id	INT	PK, FK → WaterSample	Composite primary key
polymer_id	INT	PK, FK → PolymerType	Composite primary key
particle_count	INT		CHECK ≥ 0
absorbance	DECIMAL(10,8)		
wavelength_nm	DECIMAL(6,1)		

2. Relationships

From (one side)	To (many side)	Cardinality	Line label	Crow's foot ends
Drone	Run	1 : M	operates	Exactly one ‖ to zero-or-many ○<
Run	Reading	1 : M	contains	Exactly one ‖ to one-or-many |<
SamplingPoint	Reading	1 : M	visited at	Exactly one ‖ to zero-or-many ○<
OperationalStatus	Reading	1 : M	describes	Exactly one ‖ to zero-or-many ○<
Reading	WaterSample	1 : 1 (optional)	yields	Exactly one ‖ to zero-or-one ○|
WaterSample	SampleDetection	1 : M	contains	Exactly one ‖ to zero-or-many ○<
PolymerType	SampleDetection	1 : M	found in	Exactly one ‖ to zero-or-many ○<

Many-to-many: WaterSample ↔ PolymerType. One sample can contain many polymers, and one polymer can show up in many samples. The SampleDetection junction table resolves this, and its composite key (sample_id, polymer_id) prevents duplicate rows.